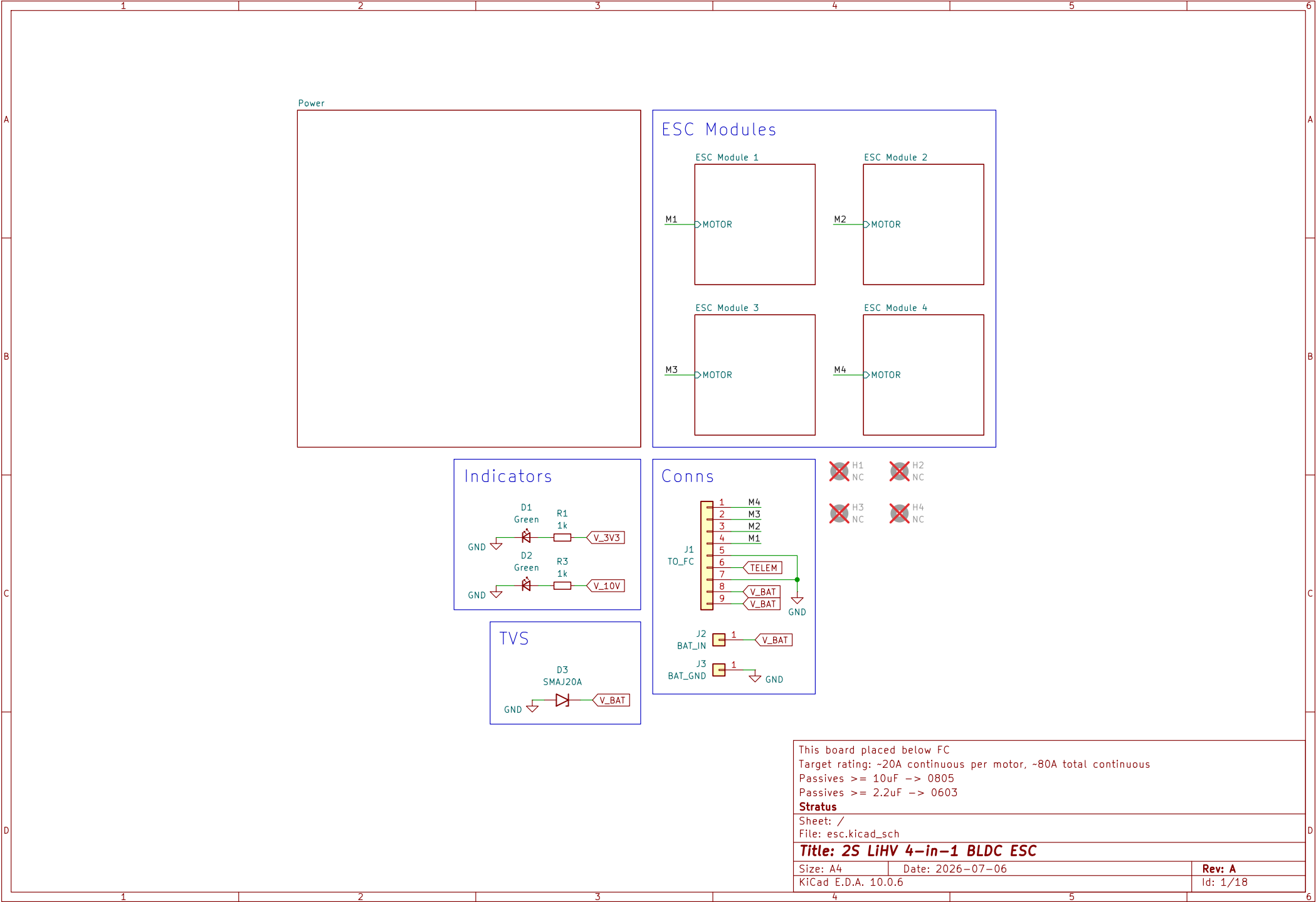
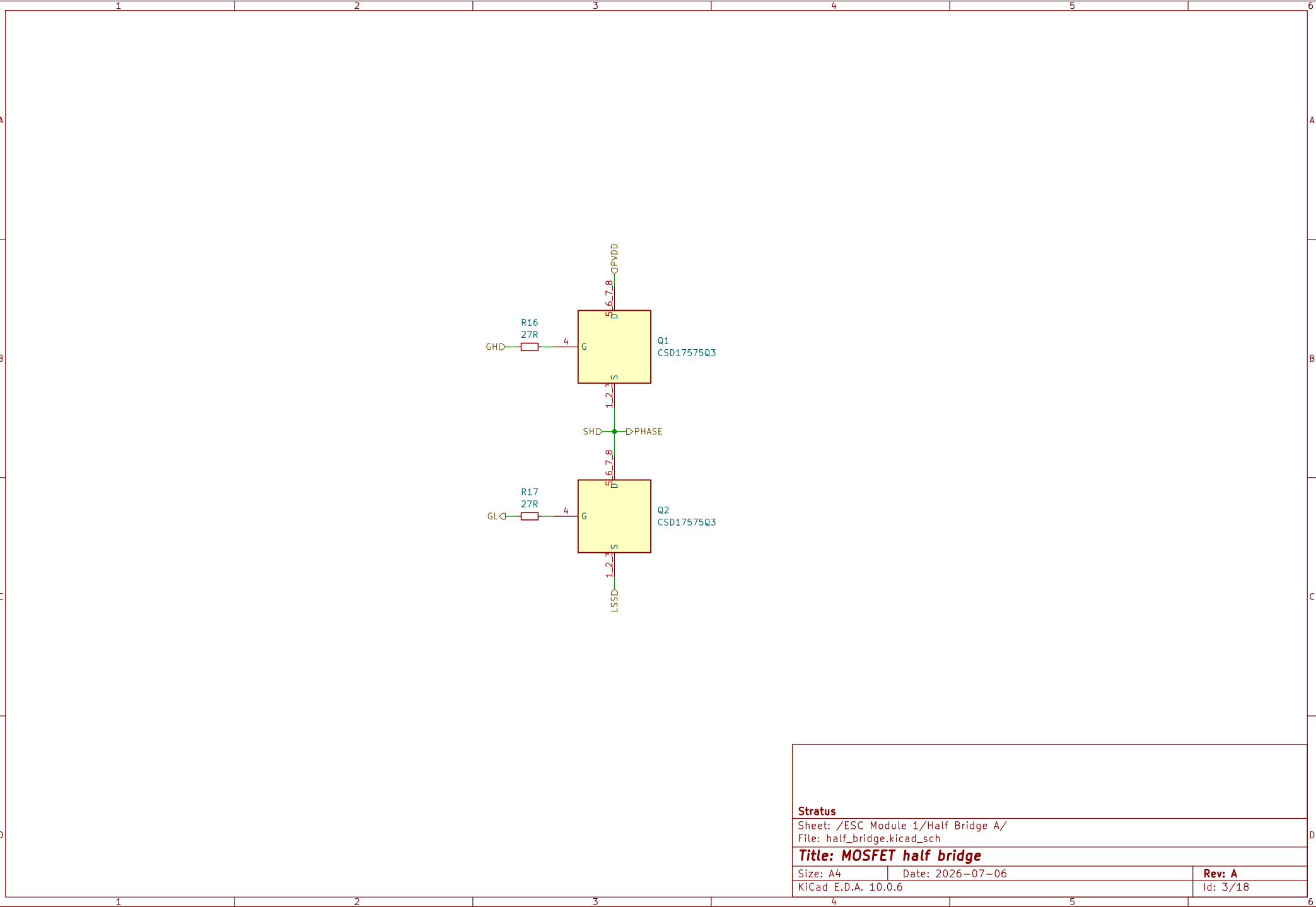
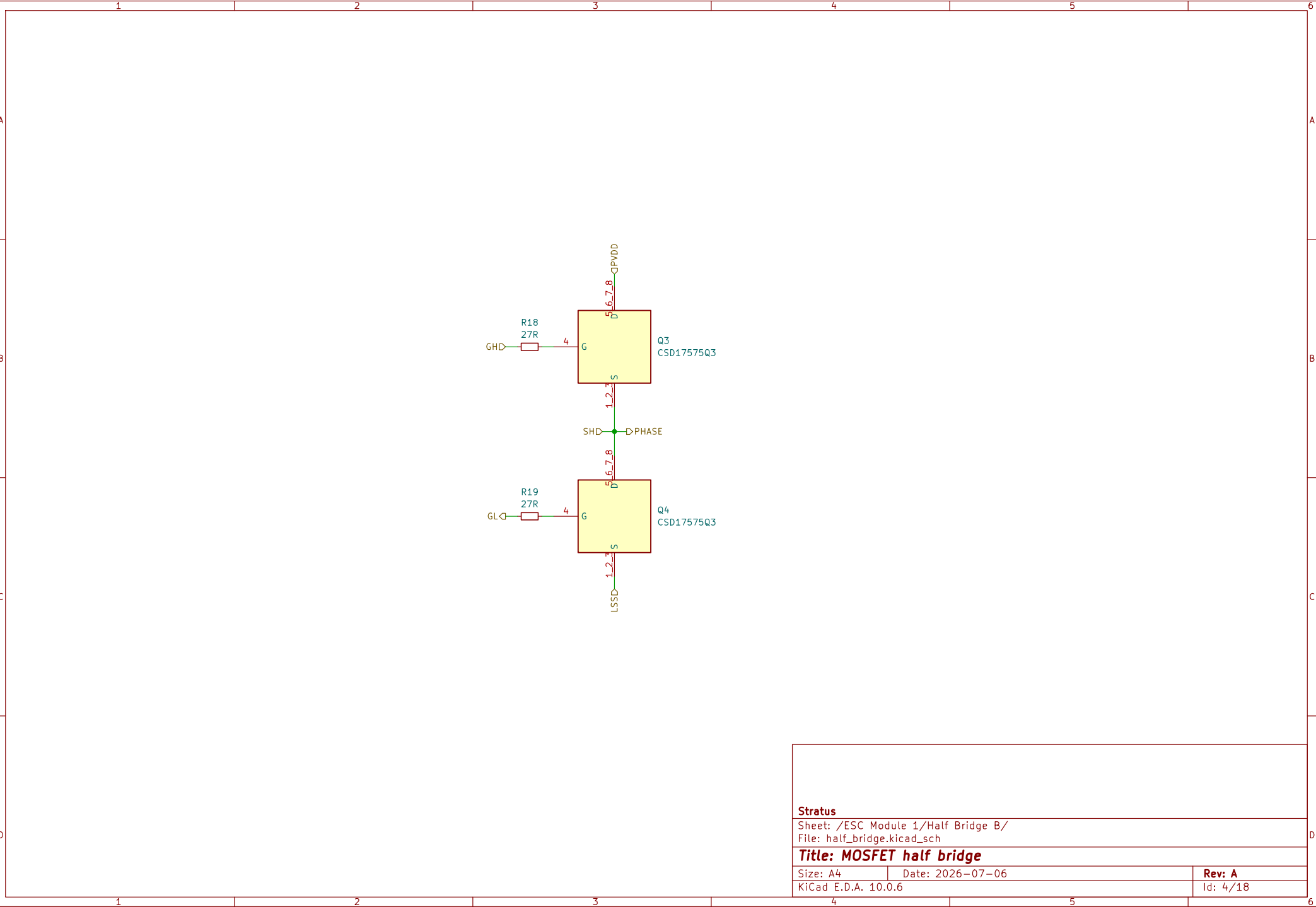






Stratus		
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File: half_bridge.kicad_sch		
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Size: A4	Date: 2026-07-06	Rev: A
KiCad E.D.A. 10.0.6		Id: 3/18

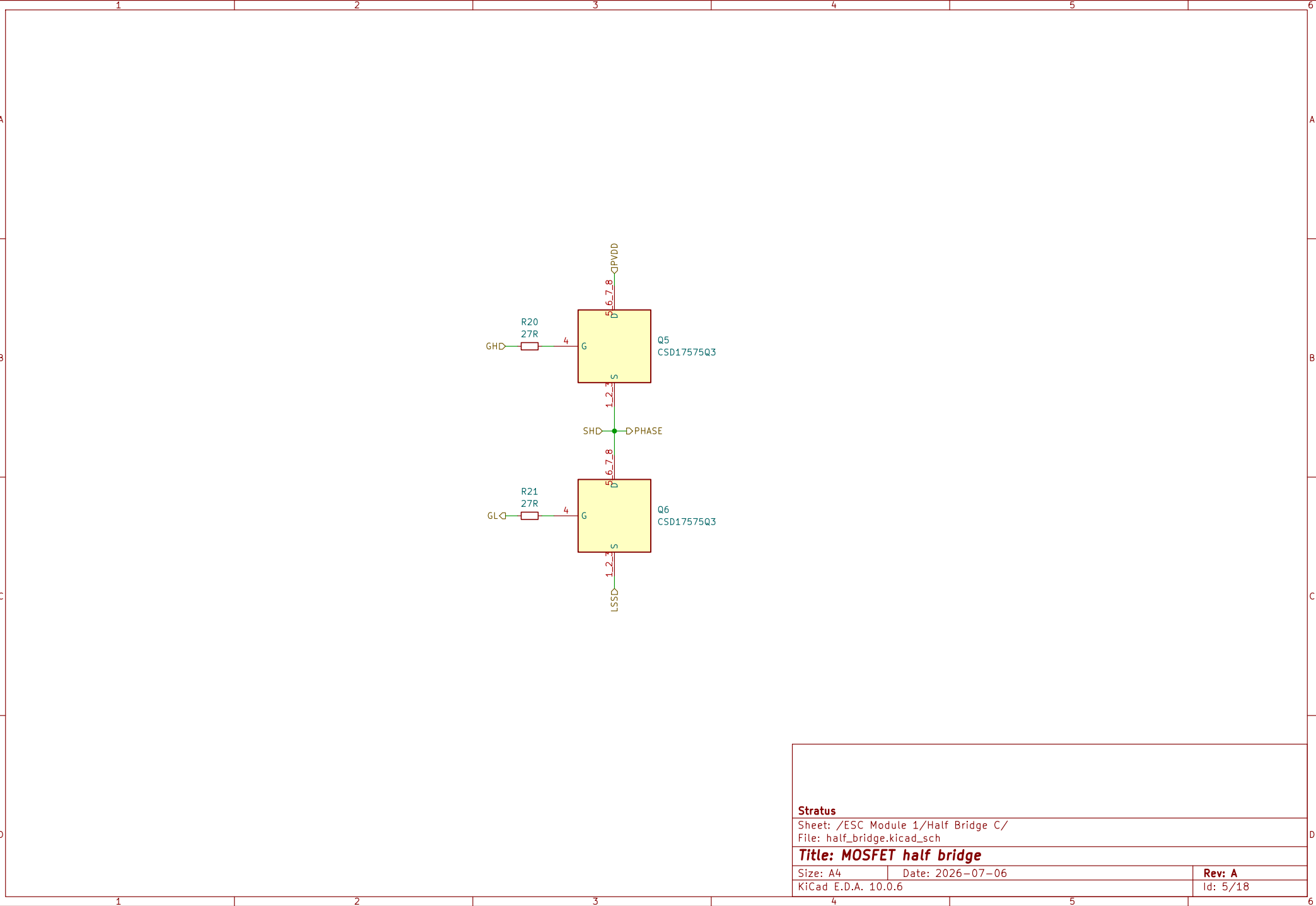


Stratus

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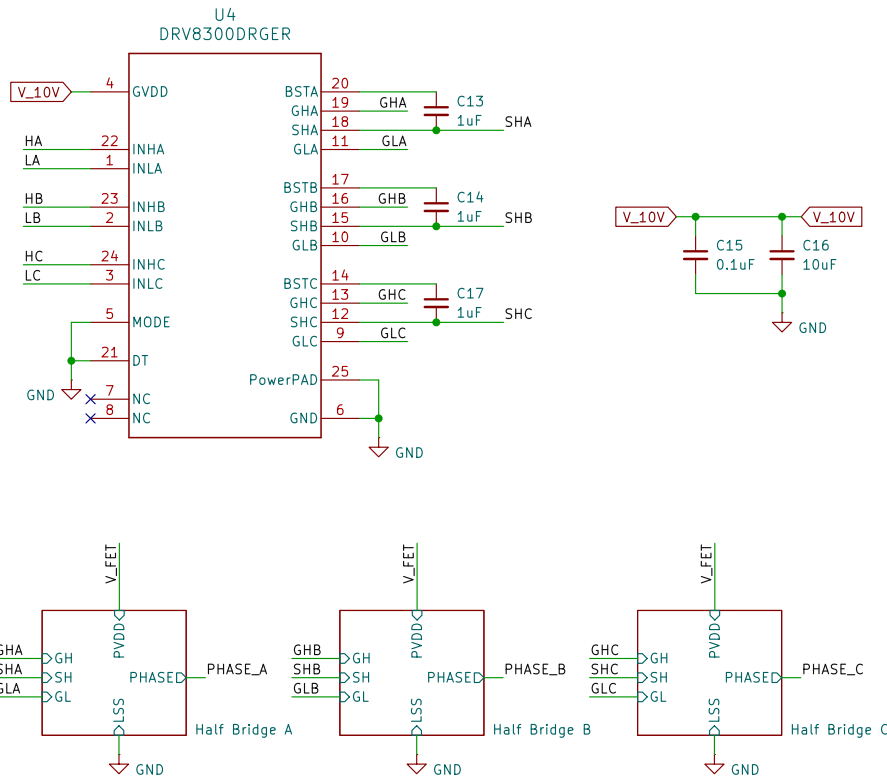
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Size: A4	Date: 2026-07-06	Rev: A
KiCad E.D.A. 10.0.6		Id: 4/18

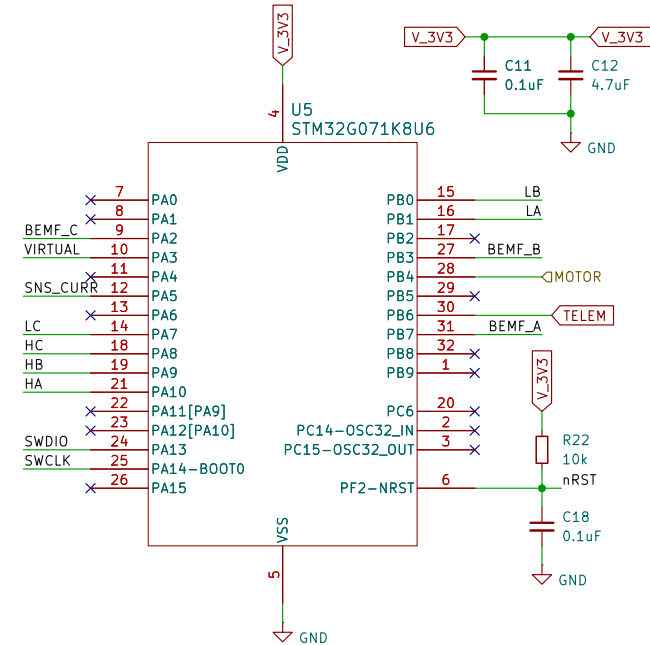


Stratus		
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Size: A4	Date: 2026-07-06	Rev: A
KiCad E.D.A. 10.0.6		Id: 5/18

Gate driver + half bridges

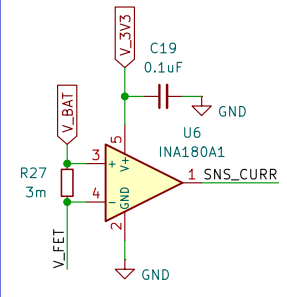


MCU

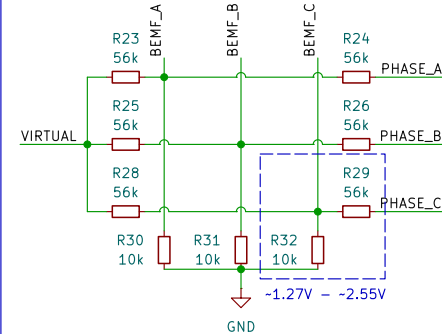


CSA

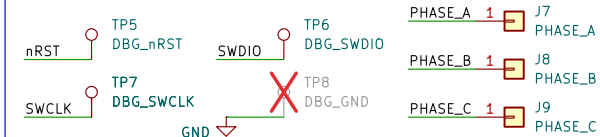
20 V/V



Back-EMF sensing

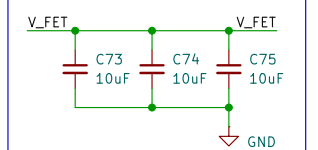


Conns



MLCCs

10uF * 3



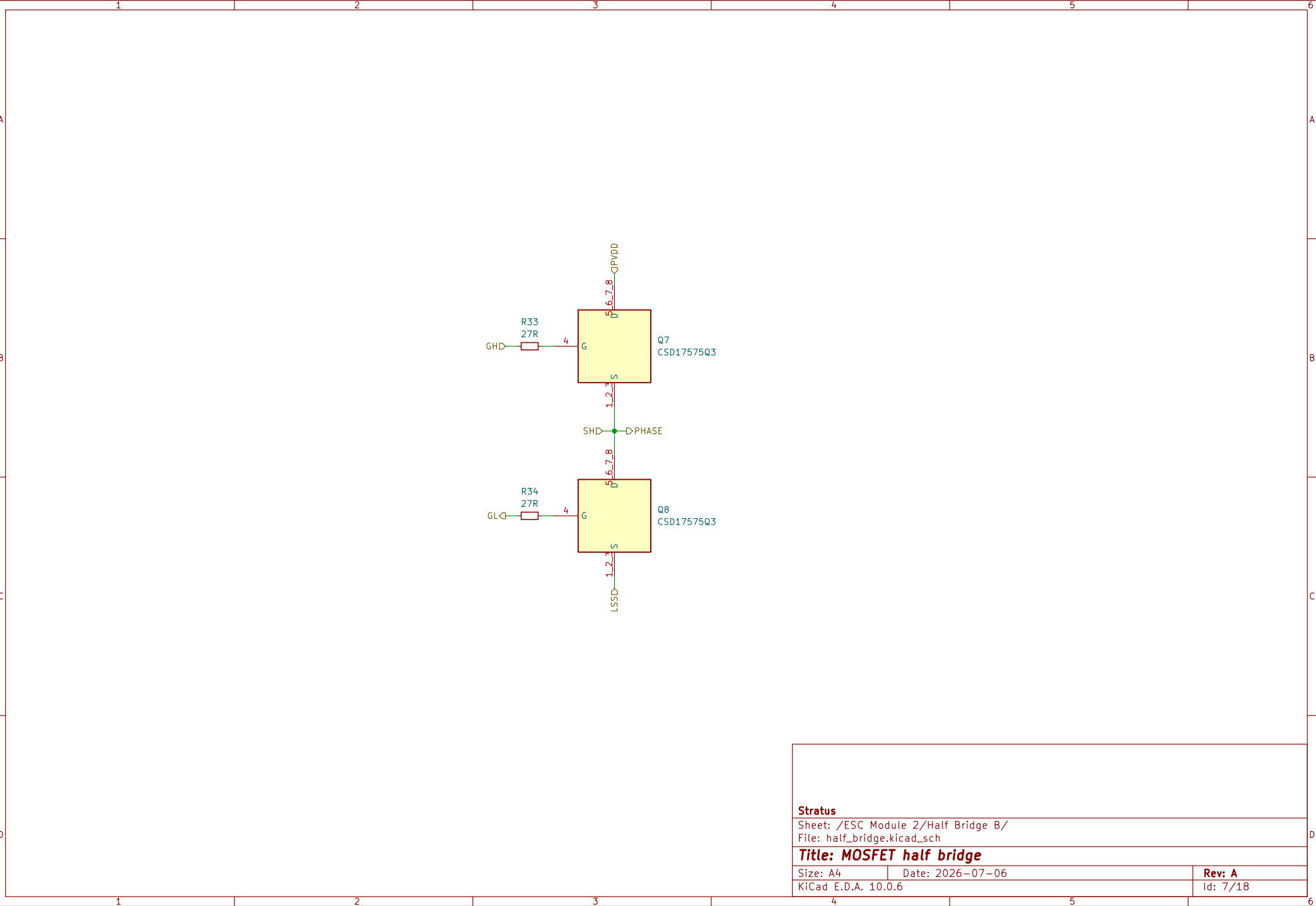
Stratus

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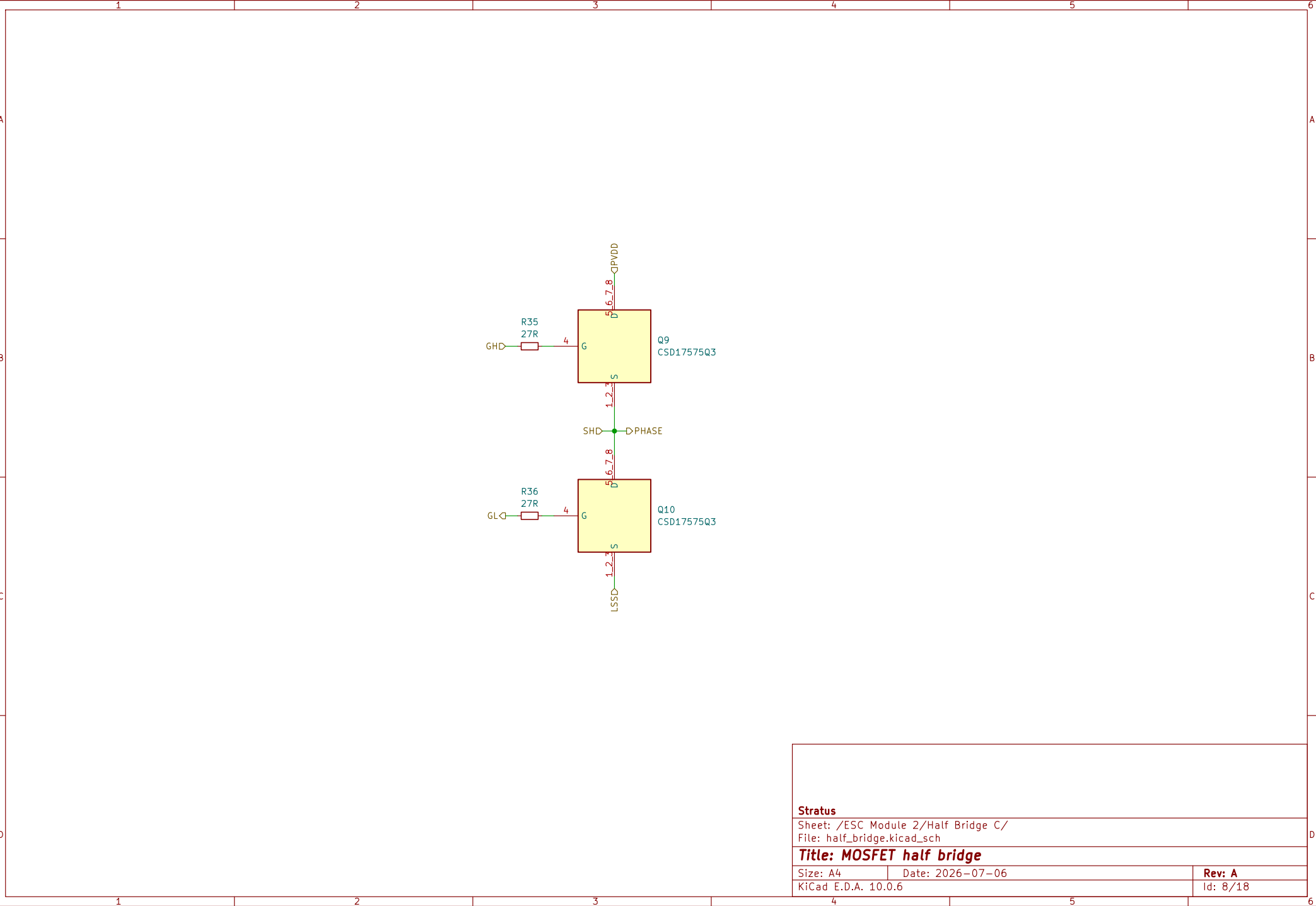
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Size: A4 Date: 2026-07-06
KiCad E.D.A. 10.0.6

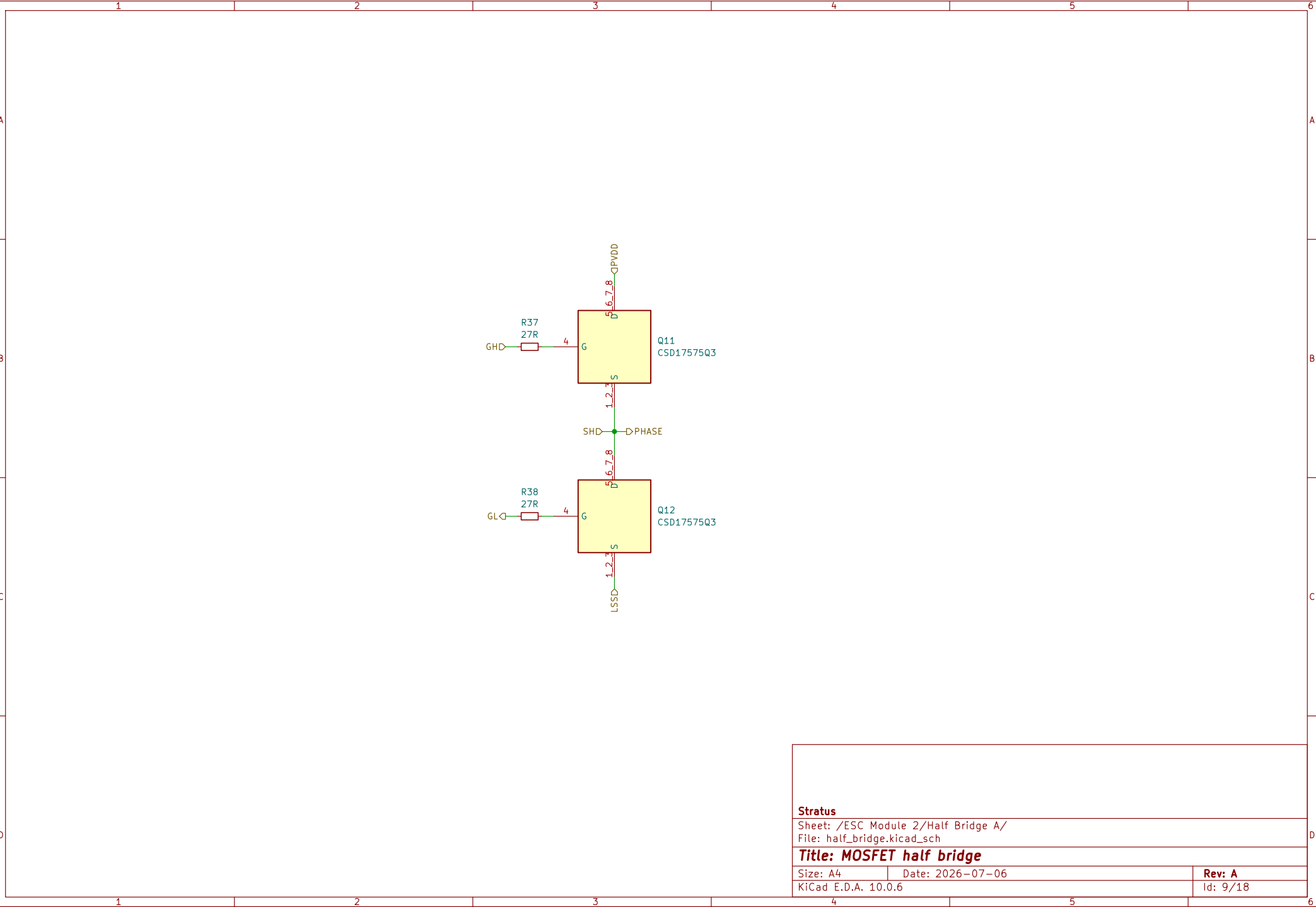
Rev: A
Id: 6/18



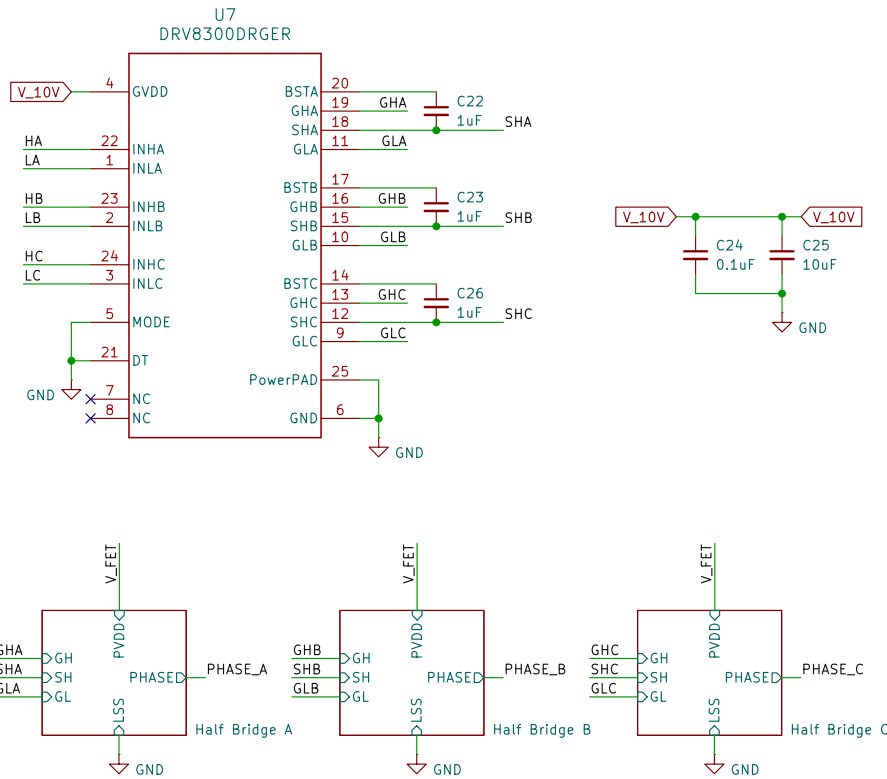
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Size: A4	Date: 2026-07-06	Rev: A
KiCad E.D.A. 10.0.6		Id: 7/18



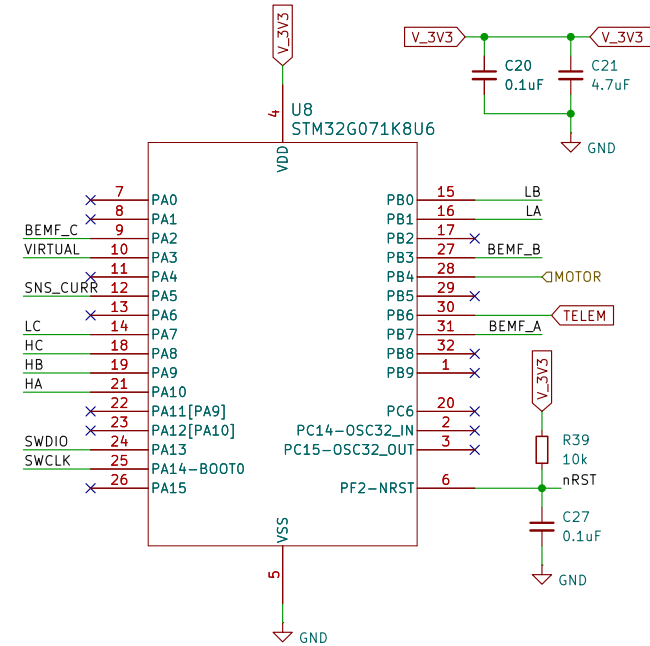
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Size: A4	Date: 2026-07-06	Rev: A
KiCad E.D.A. 10.0.6		Id: 8/18



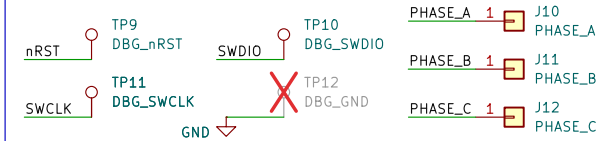
Gate driver + half bridges



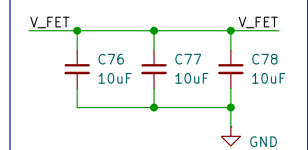
MCU



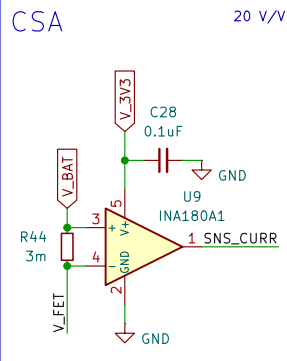
Conns



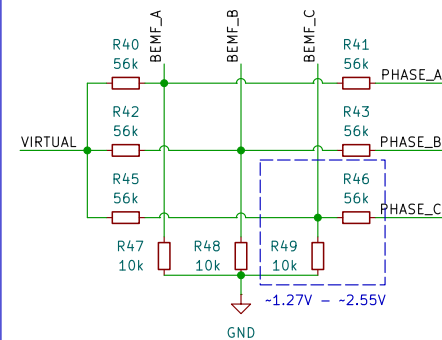
MLCCs



CSA



Back-EMF sensing



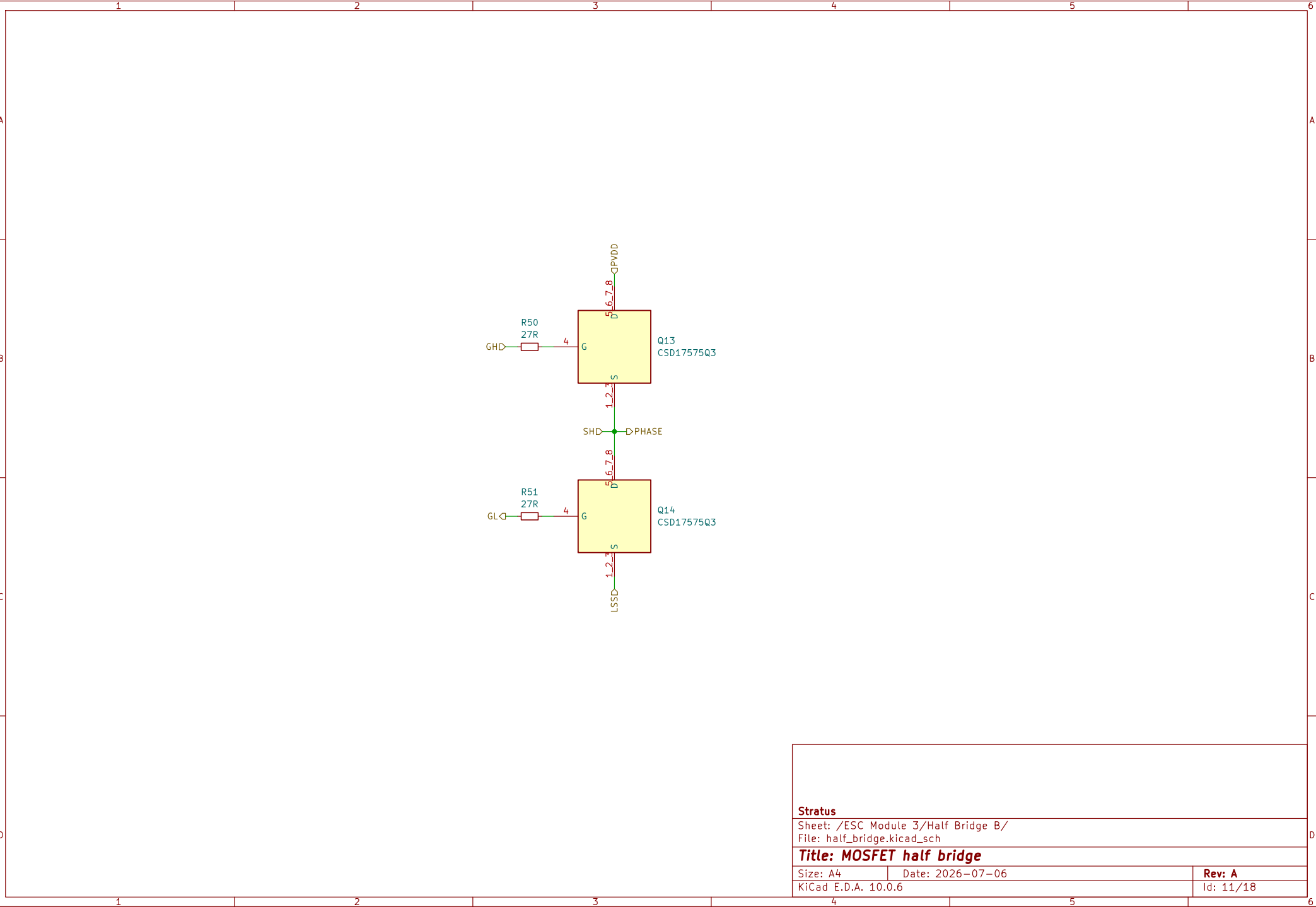
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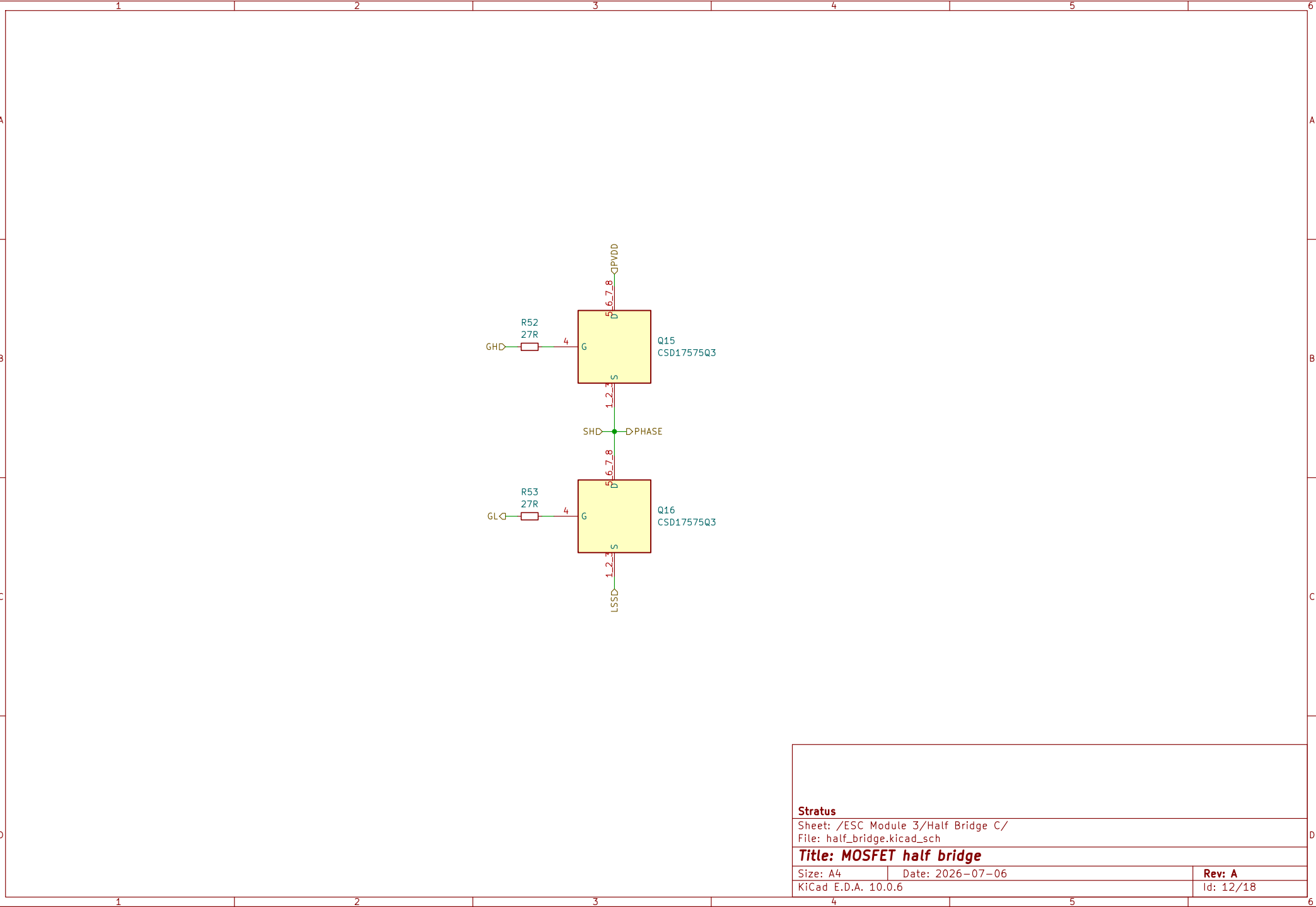
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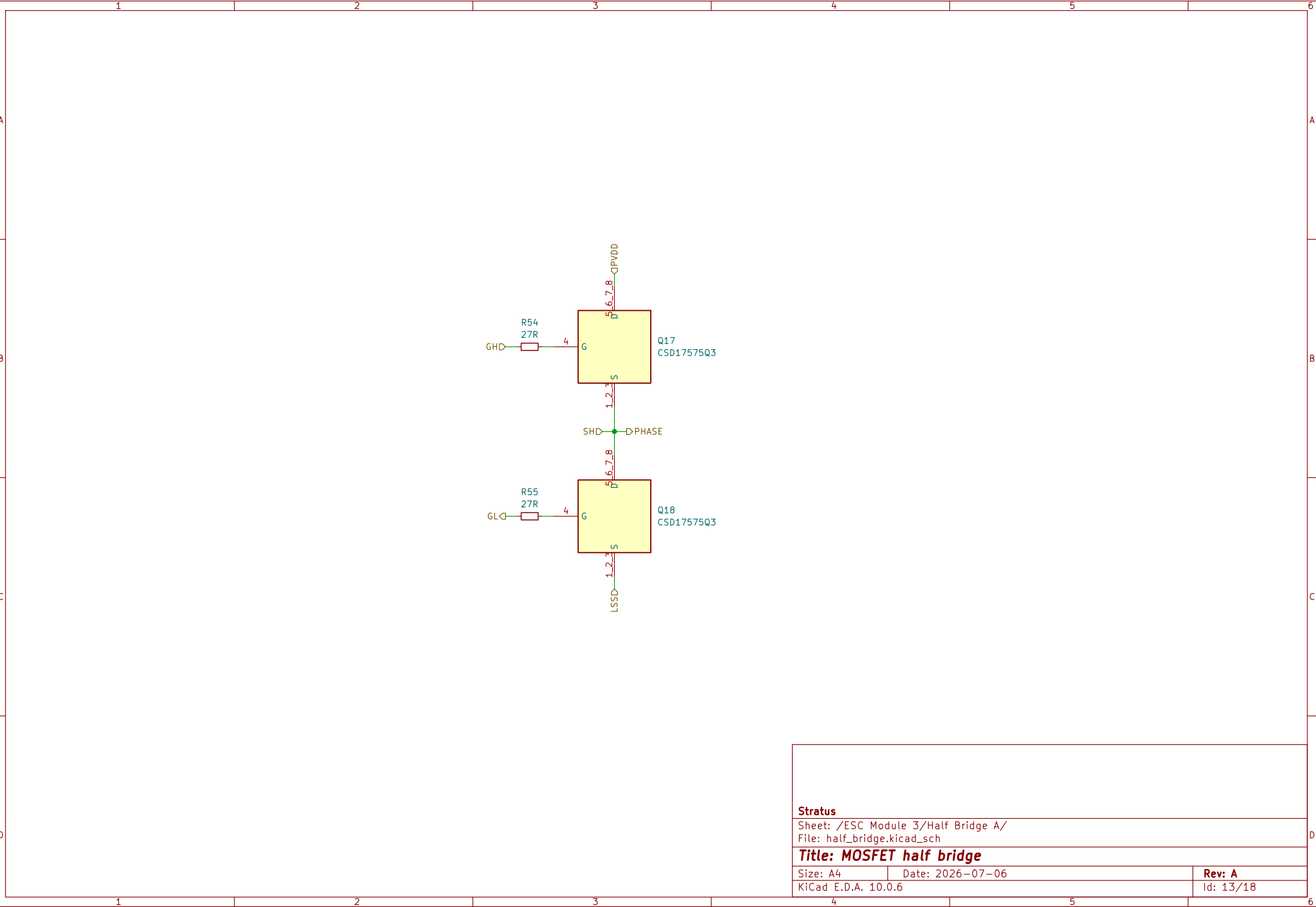
Title: ESC Submodule

Size: A4 Date: 2026-07-06
KiCad E.D.A. 10.0.6

Rev: A
Id: 10/18







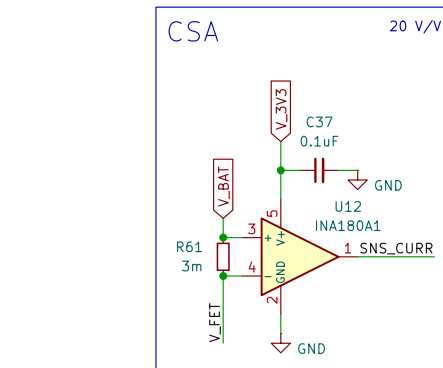
Stratus

Sheet: /ESC Module 3/Half Bridge A/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	Rev: A
KiCad E.D.A. 10.0.6		Id: 13/18

U10
DRV8300DRGER

[illegible]

The schematic diagram illustrates the connection of the STM32G071K8U6 microcontroller to various external components. The microcontroller is powered by a 3.3V supply (V_3V3) and connected to ground (GND). The connections are as follows:

- Power and Ground:** VDD is connected to V_3V3 (3.3V) and VSS is connected to GND. Decoupling capacitors C29 (0.1uF) and C30 (4.7uF) are connected between V_3V3 and GND.
- Microcontroller Pins:**
 - PA0-PA15: PA0-PA10 are connected to various sensors (BEMF_C, VIRTUAL, SNS_CURR, LC, HC, HB, HA, PA11[PA9], PA12[PA10], PA13, PA14-BOOT0, PA15). PA11 and PA12 are connected to PA9 and PA10, respectively.
 - PB0-PB9: PB0-PA9 are connected to various sensors (LB, LA, BEMF_B, MOTOR, BEMF_A, TELE). PB10 is connected to BEMF_A.
 - PC6-PC7: PC6 is connected to OSC32_IN and PC7 is connected to OSC32_OUT.
 - PF2-NRST: PF2 is connected to NRST, which is connected to a reset circuit (R56 10k, C36 0.1uF, nRST).
- Other Components:**
 - R56 (10k) and C36 (0.1uF) are connected between V_3V3 and GND.
 - nRST is connected to GND.

Pin connection diagram for the J15 connector:

- TP13 (DBG_nRST) connects to J13 PHASE_A.
- TP14 (DBG_SWDIO) connects to J14 PHASE_B.
- TP15 (DBG_SWCLK) connects to J15 PHASE_C.
- TP16 (DBG_GND) connects to GND.

A circuit diagram showing three capacitors, C79, C80, and C81, each with a value of 10uF. They are connected in parallel between a node labeled V_FET and a ground symbol labeled GND.

Sheet: /ESC Module 4/
File: esc_module.kicad_sch

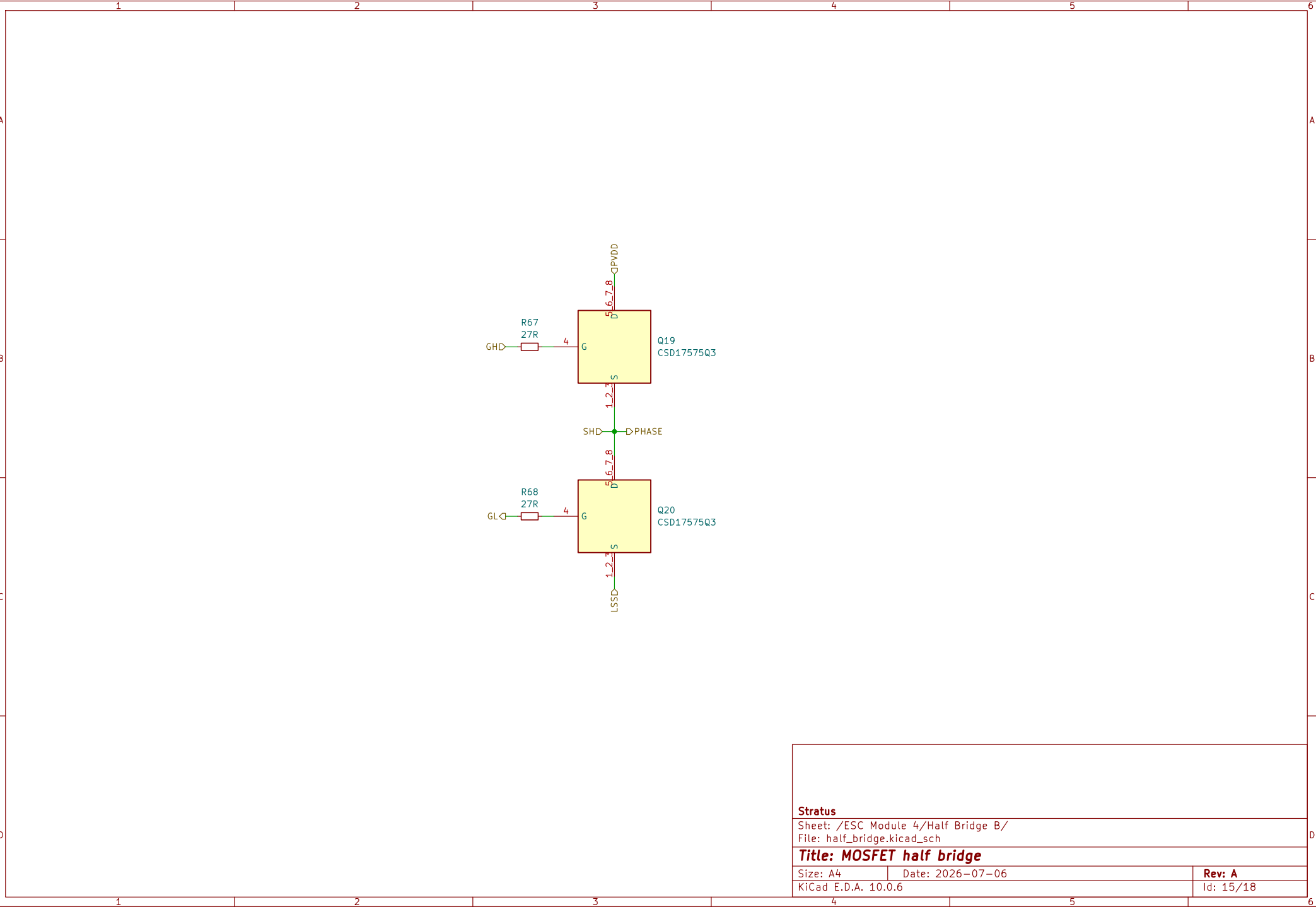
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Size: A4	Date: 2026-07-06
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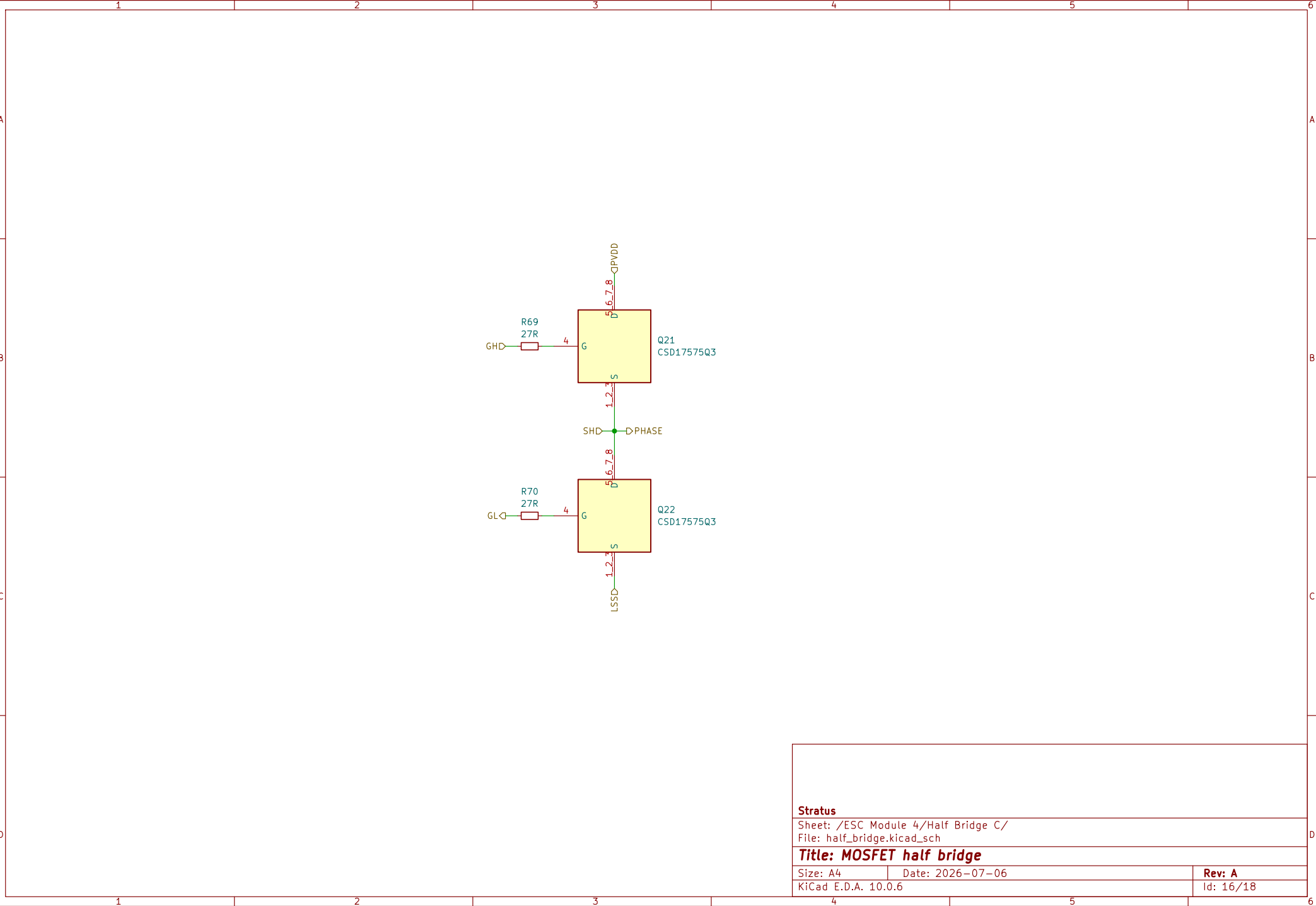
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Rev: A

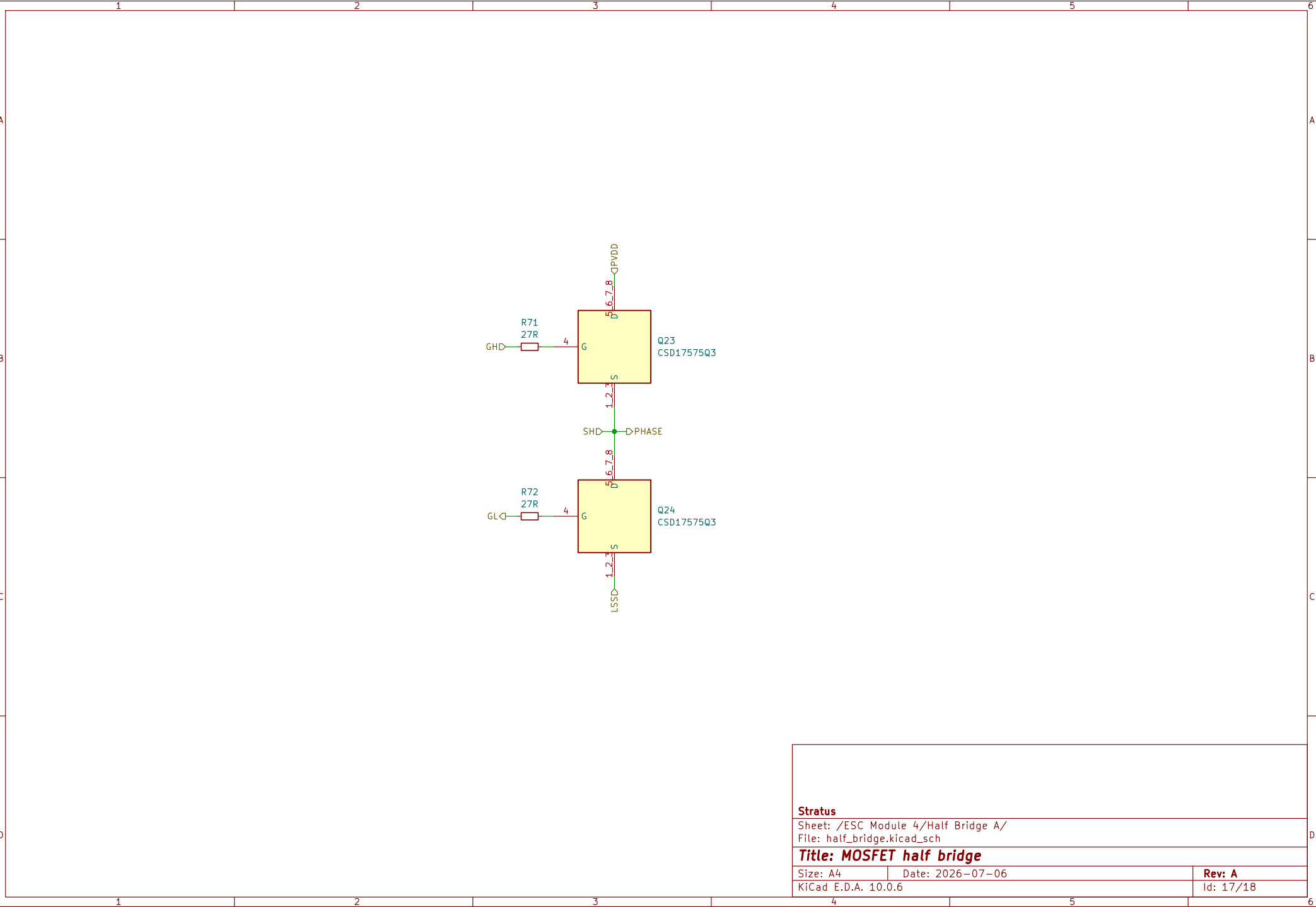
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Stratus		
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Title: MOSFET half bridge		
Size: A4	Date: 2026-07-06	Rev: A
KiCad E.D.A. 10.0.6		Id: 15/18



Stratus		
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Title: MOSFET half bridge		
Size: A4	Date: 2026-07-06	Rev: A
KiCad E.D.A. 10.0.6		Id: 16/18



Stratus		
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File: half_bridge.kicad_sch		
Title: MOSFET half bridge		
Size: A4	Date: 2026-07-06	Rev: A
KiCad E.D.A. 10.0.6		Id: 17/18

